Module V

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Testing of Hypothesis: Meaning and Significance

Hypothesis testing is an important mathematical concept that's used in the field of data science. While it's really easy to call a random method from a python library that'll carry out the test for you, it's both necessary and interesting to know what is actually happening behind the scenes! What is hypothesis testing, why do we need it, what does it do and how does it really work? Let's find out!

What is hypothesis testing?

Hypothesis testing is a statistical method to determine whether a hypothesis that you have holds true or not. The hypothesis can be with respect to two variables within a dataset, an association between two groups or a situation.

The method evaluates two mutually exclusive statements (two events that cannot occur simultaneously) to determine which statement is best supported by the sample data and make an informed decision. The building blocks of hypothesis testing are:

1. Hypotheses — null and alternate

2. Statistical tests

3. Probability distributions

4. Test statistics

5. Critical values

6. Level of significance (alpha)

7. P-value

Null and Alternative Hypotheses:

The process of hypothesis testing involves two hypotheses — a null hypothesis and an alternate hypothesis.

The null hypothesis is a statement that assumes there is no relationship between two variables, no association between two groups or no change in the current situation — hence 'null'. It is denoted by H_0.

The alternate hypothesis is the opposite of the null hypothesis because it assumes that there is some relationship between two variables or there is some change in the current situation — hence 'alternate'. It is denoted by H_a or H_1.

The conclusion of hypothesis testing is whether the null hypothesis is not rejected (so you accept that there is no change) or is rejected (so you reject that there is no change).

Let's understand this with an example.

There's a dataset that contains information on patients who've been admitted to a hospital in the past year. Two features present in this dataset are location and total amount spent. We believe there's a difference in the total amount spent during admission based on where a person lives — this statement is your alternate hypothesis, since it assumes a relationship between location and the amount spent in the hospital. Hence, the null hypothesis will be that there is no difference in the amount spent at the hospital based on where a person lives.

Now we need to figure out whether we should or should not reject the null hypothesis. How do we do that?

Statistical tests related to Testing of Hypothesis:

Statistical tests help us in rejecting or not rejecting our null hypothesis by providing us with a test statistic (a single numeric value) from our sample data. This test statistic measures the extent to which the sample data agrees with the null hypothesis — if it agrees then the null hypothesis is not rejected, else it is rejected.

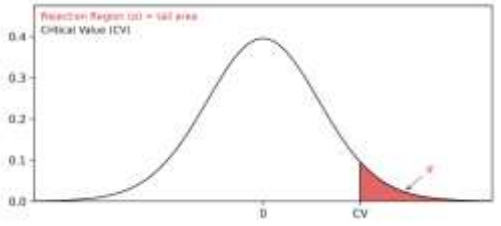
There are multiple statistical tests that exist for different use cases. The selection of which statistical test to use your hypothesis testing depends on several factors, such as — the distribution of the sample (whether it is normally distributed (follows the normal distribution)), what the sample size is, whether the variance is known, the type of data that you have, amongst some other things. Each test has its own test statistic based on the probability model/distribution that the test follows.

For example,

Test	Test Statistic	Distribution
Z-test	Z-statistic	Standard Normal Distribution
t-test	t-statistic	t-distribution
Chi-square test	Chi Square Statistic	Chi-square distribution

Statistical tests are of two types — one-tailed and two-tailed. The concept of tails is related to probability distributions. The tails of a distribution are the endings of the curve / graph on the sides of the distribution. The right end is known as the right tail or the upper tail, and the left end as the left tail or the lower tail. It is not necessary for a distribution to have both tails in all situations, it could have just one. The number of tails is dependent on your null and alternate hypotheses. So, a one-tailed statistical test is one whose distribution has only one tail — either the left (left-tailed test) or the right (right-tailed test). A two-tailed statistical test is one whose distribution has two tails — both left and right.

The purpose of a tail in statistical tests is to see whether the test statistic obtained falls within the tail or outside it. The tail region is known as the **region of rejection** — which makes sense since the null hypothesis is rejected if the test statistic falls within this area. If it falls outside the region, then the null hypothesis is accepted.



Determining where the tail begins is done with the help of a parameter known as the **level of significance** — alpha. It is the probability of rejecting the null hypothesis when the null hypothesis is true. The most common value for alpha is 5%, i.e. 0.05. This means that you face a 5% risk of rejecting the null hypothesis, i.e. believing there is a difference in the current situation when there actually isn't.

In terms of our distribution, alpha is the area of the distribution that makes up the region of rejection — hence it is the tail(s). Since the tails are at the end of the distribution, we are literally saying, with the level of significance, how far away from the null hypothesis our test statistic needs to be for us to actually reject the null hypothesis.

So, if we were to select a right-tailed test and our level of significance was 5%, then the 5% of the area of distribution on the right end of the curve would be our region of rejection. Using this, we calculate the value on the x-axis which demarcates where the region of rejection begins — this is known as the critical value. In the case of a two-tailed test, we would divide the level of significance by half — 2.5% — and this would be the area of each region of rejection on either

side of the curve. Then we would have two critical values and the regions of rejection would also be known as the critical regions.

One-tailed tests are directional as there is only one region of rejection. You are looking for a test statistic that falls within that sole region of rejection. Two-tailed tests are nondirectional since there are two regions of rejection and if the test statistic falls in either of them then the null hypothesis is rejected.

For example, the sample mean age of patients at a dentist's clinic is 18 — this is your null hypothesis (since this is the current scenario and you are assuming there is no change). Another dentist joins the practice and you believe that now the sample mean of patients will be greater than 18 — this is your alternate hypothesis. It is directional due to the 'greater than' — only if the sample mean is greater than 18 will you reject the null hypothesis. If it is less than 18, your null hypothesis will not be rejected.

If it were a two-tailed test, your alternate hypothesis would simply be that the sample mean age of patients is not equal to 18. The sample mean could be greater than or less than 18, either way the null hypothesis would be rejected, hence it is nondirectional.

So far, we have hypotheses, a statistical test based on a probability distribution, a level of significance from which we get the region of rejection from which we get the critical value, and a test statistic that falls somewhere within our distribution. But, where does the **p-value** come in?

As mentioned before, the test statistic is compared with the critical value. The p-value is compared with the level of significance.

P-value is the probability that the test statistic you have obtained is by random chance which would mean the null hypothesis is actually true. If the p-value is lower than the level of significance then it is statistically significant and the null hypothesis is rejected, because it means that there is a less than 5% probability that the null hypothesis is true. If it is higher than the level of significance then the null hypothesis is not rejected.

A big plus point of the p-value is that it doesn't need to be recalculated for different levels of significance since it is directly comparable. In the case of test statistics, they have to be compared with critical values which do need to be recalculated for different levels of significance. That's why it's always better to work with the p-value in hypothesis testing.

Despite the above-mentioned advantage though, the p-value is still probabilistic. Hence there is a chance that the null hypothesis will be rejected even though it is true. Alpha (the level of

significance) is the likelihood that this will happen. This error is known as a **type I error** or a false positive.

The opposite to the above error would be not rejecting the null hypothesis even though it is not true. This is known as a **type II error** or a false negative.

For example, in a court of law where you are "innocent until proven guilty," unfortunately, sometimes innocent people are convicted when they shouldn't have been, and guilty people are let free when they, in fact, did the crime. These two mistakes correspond to what are known in statistical testing as Type I and Type II errors, respectively.

		Truth about the null hypothesis (Actual)	
		True	False
Decision About the null hypothesis (Predicted)	Accept	Correct Decision	Type-II error
	Reject	Type-I error	Correct Decision

Type I and Type II errors are inversely related to each other, as reducing one results in increasing the other. In different scenarios, one type of error can be more detrimental and therefore more important to minimize. For example, in the court situation, it is better to let a guilty person walk than it would be to send an innocent person to jail. Therefore, in this situation a Type I error would be worse. In medical situations, however, it is better to have a positive test result for something you don't have than to have a negative test result for something you do have. By testing negative for a disease, you actually have, your life can be put in jeopardy due to lack of treatment. Thus, in this situation a Type II error would be worse. While completely eliminating the possibility of these errors is impossible for non-deterministic equations, minimizing them is a big goal in statistics and data science.



One-sided hypothesis

A **one-sided** alternative hypothesis to determine whether the population parameter differs from the hypothesized value in a specific direction.

One-sided tests are two types:

(1) left-tailed (or, lower tail) test, and (2) right-tailed (or, upper tail) test.

For, example,

Null Hypothesis, $H_0 : \mu = 98.6$

Alternative Hypothesis, $H_1 : \mu < 98.6$

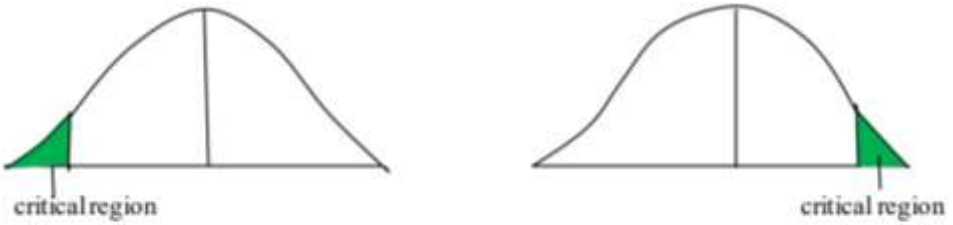
is an example of left-tailed test.

And, Null Hypothesis, $H_0 : \mu = 98.6$

Alternative Hypothesis, $H_1 : \mu > 98.6$

is an example of right-tailed test.

The following diagram is an example of the rejection region of a one-sided test.



Left-tailed test Right-tailed test

Two-sided hypothesis

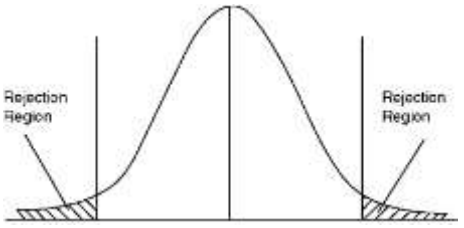
A **two-sided (or, two-tailed)** alternative hypothesis to determine whether the population parameter is either greater than or less than the hypothesized value.

For example,

Null Hypothesis, $H_0 : \mu = 98.6$

Alternative Hypothesis, $H_1 : \mu \neq 98.6$

The following diagram is an example of the rejection region of a two-sided test.



Finally, we have all the building blocks of hypothesis testing sorted so the process is also pretty clear. The steps for hypothesis testing are:

1. Formulating your hypotheses

2. Choosing a level of significance

3. Identifying a statistical test

4. Calculating the test statistic

5. Calculating the p-value and comparing it with the level of significance

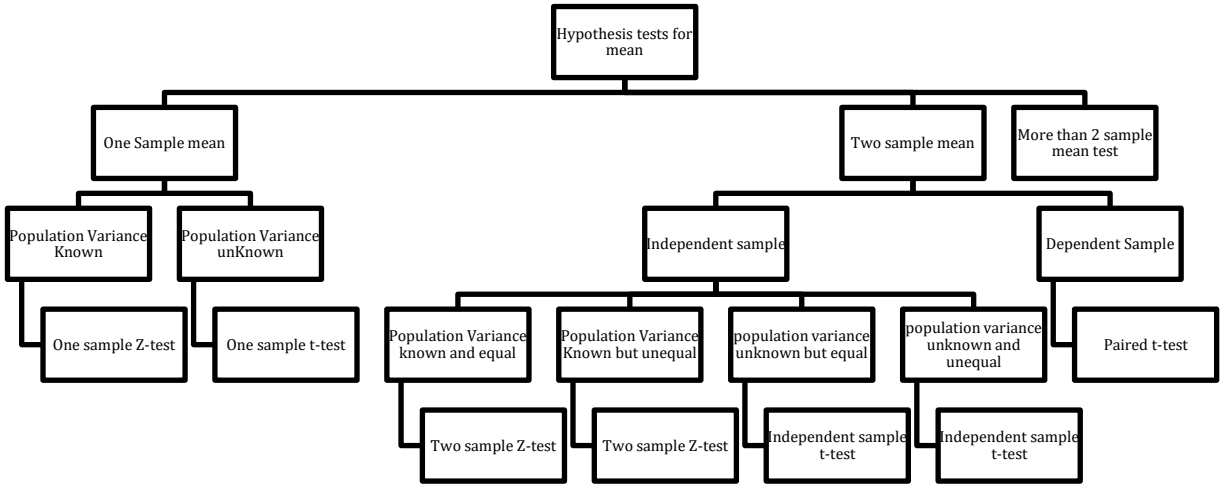
6. Concluding — rejecting or not rejecting your null hypothesis.

Testing of Hypothesis for mean

Having collected together a number of tests, it is necessary to consider what can be tested, and we include here some very general remarks about the general problem of hypothesis testing. Broadly speaking there are two basic concepts to grasp before commencing. First, the tests are designed neither to prove nor to disprove hypotheses. We never set out to prove anything; our aim is to show that an idea is untenable as it leads to an unsatisfactorily small probability. Second, the hypothesis we are trying to disprove is always chosen to be the one in which there is no change; for example, there is no difference between the two population means, between the two samples, etc. This is why it is usually referred to as the null hypothesis.

Most of the parametric tests considered in this chapter have made the assumption that the populations involved have normal distributions. Therefore, a test should only be carried out when the normality assumption is not violated. It is also a necessary part of the test to check the effect

of applying these tests when the assumption of normality is violated. However, we'll discuss different hypothesis tests on population mean. The hypothesis testing for mean can be classified in these ways:



Z-test for one population mean (when population variance is known):

Objective:

To investigate the significance of the difference between an assumed population mean μ_0 and a sample mean $\bar{X}$.

Limitations:

1. It is necessary that the population variance σ^2 is known. If σ^2 is not known, then, the t-test for a population mean will be more appropriate.

2. The test is accurate if the population is normally distributed. If the population is not normal, the test will still give an approximate guide.

Method:

From a population with assumed mean μ_0 and known variance σ^2, a random sample of size n is taken and the sample mean $\bar{X}$ is calculated. The test statistic

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

may be compared with the standard normal distribution using either a one-tailed or two-tailed test, with critical region of size α.

Example:

For a particular range of cosmetics, a filling process is set to fill tubs of face powder with 4 gm on average and standard deviation 1 gm. A quality inspector takes a random sample of nine tubs and weighs the powder in each. The average weight of powder is 4.6 gm. What can be said about the filling process?

Answer:

A two-tailed test is used if we are concerned about over- and under-filling.

At 5% level of significance, the critical value of z-table for one-tailed is 1.645 and the critical value of z-table for two-tailed is 1.96.

Using the given data, we can find Z = 1.8 and if our acceptance range is $-1.96 < Z < 1.96$, so we do not reject the null hypothesis. That is, there is no reason to suggest, for this sample, that the filling process is not running on target.

On the other hand, if we are only concerned about over-filling of the cosmetic then a one-tailed test is appropriate. The acceptance region is now $Z < 1.645$. Notice that we have fixed our probability, which determines our acceptance or rejection of the null hypothesis, at 0.05 (or 10 per cent) whether the test is one- or two-tailed. So now we reject the null hypothesis and can reasonably suspect that we are over-filling the tubs with cosmetic.

Quality control inspectors would normally take regular small samples to detect the departure of a process from its target, but the basis of this process is essentially that suggested above.

t-test for one population mean (when population variance is unknown):

Objective:

To investigate the significance of the difference between an assumed population mean μ_0 and a sample mean $\bar{X}$.

Limitations:

1. If the variance of the population σ^2 is known, a more powerful test is available: the Z-test for a population mean.

2. The test is accurate if the population is normally distributed. If the population is not normal, the test will give an approximate guide.

Method:

From a population with assumed mean μ_0 and unknown variance, a random sample of size n is taken and the sample mean is calculated as well as the sample standard deviation using the formula

$$s = \left\{ \frac{\sum_{i=1}^{n} (x_i - \bar{x})}{n-1} \right\}^{\frac{1}{2}}$$

The test statistic is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

which may be compared with Student's t-distribution with n − 1 degrees of freedom. The test may be either one-tailed or two-tailed.

Example:

A sample of nine plastic nuts yielded an average diameter of 3.1 cm with sample standard deviation of 1.0 cm. It is assumed from design and manufacturing requirements that the population mean of nuts is 4.0 cm. What does this say about the mean diameter of plastic nuts being produced?

Answer:

Since we are concerned about both under- and over-sized nuts (for different reasons) a two-tailed test is appropriate.

The critical value of t-table for 5% level of significance and 8 degrees of freedom is 2.3.

Using the formula, our computed t value is −2.7 and acceptance region $-2.3 < t < 2.3$. We reject the null hypothesis and accept the alternative hypothesis of a difference between the sample and population means. There is a significant difference (a drop-in fact) in the mean diameters of plastic nuts (i.e., between the sample and population)

Important Tables:

Critical values table for a standard normal distribution:

Level of significance α		z
Two-sided	One-sided	
0.001	0.0005	3.29
0.002	0.001	3.09
0.0026	0.0013	3.00
0.01	0.005	2.58
0.02	0.01	2.33
0.0456	0.0228	2.00
0.05	0.025	1.96
0.10	0.05	1.64
0.20	0.10	1.28
0.318	0.159	1.00

Critical values table of the t-distribution

	Level of significance α				
ν	0.10	0.05	0.025	0.01	0.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169

11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
∞	1.282	1.645	1.960	2.326	2.576

Z-test for two population means (variances known and equal)

Objective:

To investigate the significance of the difference between the means of two populations.

Limitations:

1. Both populations must have equal variances and this variance σ^2 must be known. (If σ^2 is not known, then the t-test for two population means will be appropriate.)

2. The test is accurate if the populations are normally distributed. If not normal, the test may be regarded as approximate.

Method:

Consider two populations with means μ_1 and μ_2. Independent random samples of size n_1 and n_2 are taken which give sample means $\overline{x}_1$ and $\overline{x}_2$. The test statistic

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma\left(\frac{1}{n_1} + \frac{1}{n_2}\right)^{\frac{1}{2}}}$$

may be compared with the standard normal distribution using either a one-tailed test or two-tailed test.

Example:

Two teams of financial sales persons are compared to see if it is likely that the instruction each has received could have led to differing success rates. A sample of nine transactions (which involves the whole team) yields an average success rate of 1.2. Similarly, a sample of 16 transactions for the second team yields a success rate of 1.7. The variances for both teams are equal to 2.0750 (standard deviation 1.4405). The success rate is calculated using a range of output measures for a transaction.

If we are only interested to know of a difference between the two teams then a two-tailed test is appropriate. In this case we accept the null hypothesis and can assume that both teams are equally successful. This is because our acceptance region is $-1.96 < Z < 1.96$ and we have computed a Z value, for this sample, of -0.833.

On the other hand, if we suspect that the first team had received better training than the second team, we would use a one-tailed test.

For our example, here, this is certainly not the case since our Z value is negative. Our acceptance region is $Z < 1.645$. Since the performance is in the wrong direction, we don't even need to perform a calculation. Notice that we are not doing all possible combination of tests so that we can find a significant result. Our test is based on our design of the 'experiment' or survey planned before we collect any data. Our data do not have a bearing on the form of the testing.

Z-test for two population means (variances known and unequal)

Objective:

To investigate the significance of the difference between the means of two populations.

Limitations:

1. Both populations must have unequal variances and the variances of two population must be known. (If the variances are not known, then the t-test for two population means will be appropriate.)

2. The test is accurate if the populations are normally distributed. If not normal, the test may be regarded as approximate.

Method:

Consider two populations with means μ_1 and μ_2 with variances σ_1^2 and σ_2^2. Independent random samples of size n_1 and n_2 are taken which give sample means $\bar{x}_1$ and $\bar{x}_2$. The test statistic

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)^{\frac{1}{2}}}$$

may be compared with the standard normal distribution using either a one-tailed or two-tailed test.

Example:

Brand A of a jumbo-sized pack of potato crisp is known to have a more variable weight than brand B of potato crisp. Population variances are 0.000576 gm2 and 0.001089 gm2, respectively. The respective means for samples of size 13 and 8 are 80.02 gm and 79.98 gm. Is there a difference between the two brands in terms of the weights of the jumbo packs?

We do not have any pre-conceived notion of which brand might be ‘heavier’ so we use a two-tailed test. Our acceptance region is $-1.96 < Z < 1.96$ and our calculated Z value of 2.98. We therefore reject our null hypothesis and can conclude that there is a difference with brand B yielding a heavier pack of crisps.

t-test for two population means (variances are unknown but equal)

Object:

To investigate the significance of the difference between the means of two populations.

Limitations:

1. If the variance of the populations is known, a more powerful test is available: the Z-test for two population means.

2. The test is accurate if the populations are normally distributed. If the populations are not normal, the test will give an approximate guide.

Method

Consider two populations with means μ_1 and μ_2. Independent random samples of size n_1 and n_2 are taken from which sample means $\bar{x}_1$ and $\bar{x}_2$ together with sums of squares

$s_1^2 = \sum_{i=1}^{n_1}(x_i - \bar{x}_1)^2$, and, $s_2^2 = \sum_{i=1}^{n_2}(x_i - \bar{x}_2)^2$

Are calculated.

The best estimate of the population variance is found as pooled variance,

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

The test statistic is

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s\left(\frac{1}{n_1} + \frac{1}{n_2}\right)^{\frac{1}{2}}}$$

which may be compared with Student's t-distribution with $n_1 + n_2 - 2$ degrees of freedom. The test may be either one-tailed or two-tailed.

Example:

Two snack foods are made and sold in 30 gm packets. Random samples of size 12 are taken from the production line of each snack food and means and variances obtained viz.: mean-1= 31.75, variance-1= 112.25; mean-2= 28.67, variance-2= 66.64. What can be said about the two production processes in relation to the weight of packets?

We use a two-tailed test and find that t is 0.798. Our acceptance region is $-2.07 < t < 2.07$ and so we accept our null hypothesis. So, we can conclude that the mean weight of packs from the two production lines is the same.

t-test for two population means (variances unknown and unequal):

Objective:

To investigate the significance of the difference between the means of two populations.

Limitations:

1. If the variances of the populations are known, a more powerful test is available: the Z-test for two population means.

2. The test is approximate if the populations are normally distributed or if the sample sizes are sufficiently large.

3. The test should only be used to test the hypothesis $\mu_1 = \mu_2$.

Method:

Consider two populations with means μ_1 and μ_2. Independent random samples of size n_1 and n_2 are taken from which sample means $\bar{x}_1$ and $\bar{x}_2$, and, variances

$s_1^2 = \dfrac{\sum_{i=1}^{n_1}(x_i - \bar{x}_1)^2}{n_1 - 1}$, and, $s_2^2 = \dfrac{\sum_{i=1}^{n_2}(x_i - \bar{x}_2)^2}{n_2 - 1}$ are calculated.

The test statistic is

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\left(\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}\right)^{\frac{1}{2}}}$$

which may be compared with Student's t-distribution with degrees of freedom given by

$$\nu = \left\{ \frac{\left\{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}\right\}^2}{\dfrac{s_1^4}{n_1^2(n_1 - 1)} + \dfrac{s_2^4}{n_2^2(n_2 - 1)}} \right\}$$

Example:

Two financial organizations are about to merge and, as part of the rationalization process, some consideration is to be made of service duplication. Two sales teams responsible for essentially identical products are compared by selecting samples from each and reviewing their respective profit contribution levels per employee over a period of two weeks. These are found to be 3166.00 and 2240.40 with estimated variance of 6328.27 and 221 661.3 respectively. How do the two teams compare on performance?

We compute a t value of 5.72. Our acceptance region is $-2.26 < t < 2.26$ so we reject the null hypothesis and accept the alternative. There is a significant difference between the two teams. Team 1 is more productive than team 2.

t-test for two population means (method of paired comparisons)

Object:

To investigate the significance of the difference between two population means. No assumption is made about the population variances.

Limitations:

1. The observations for the two samples must be obtained in pairs. Apart from population differences, the observations in each pair should be carried out under identical, or almost identical, conditions.

2. The test is accurate if the populations are normally distributed. If not normal, the test may be regarded as approximate.

Method:

The differences d_i are formed for each pair of observations. If there are n such pairs of observations, we can calculate the variance of the differences by

$$s^2 = \sum_{i=1}^{n} \frac{(d_i - \bar{d})^2}{n-1}$$

Let the means of the samples from the two populations be denoted by $\bar{x}_1$ and $\bar{x}_2$. Then the test statistic becomes

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{s/n^{\frac{1}{2}}}$$

which follows Student's t-distribution with n − 1 degrees of freedom. The test may be either one-tailed or two-tailed.

Example:

To compare the efficacy of two treatments for a respiratory condition, ten patients are selected at random and the treatments are administered using an oral spray. The patients then perform a treadmill exercise until a maximum exercise rate is reached. The times for these are compared. A suitable period of time is ensured between treatments to prevent the effect of treatments to interact. Do the two treatments differ?

In this case we do not expect one particular treatment to be superior to the other so a two-tailed test is used. We compute a t value of −0.11 and have an acceptance region of $-2.26 < t < 2.26$. So, we accept the null hypothesis of no difference between the two treatments. However, in such situations it is often the case that an improvement over an existing or original treatment is expected. Then a one-tailed test would be appropriate.

QUESTIONS:

Multiple Choice Questions:

1.	Identify which of the following distributions used in ANOVA tests	
	a. F-test	b. T-test
	c. Z-test	d. Chi-square
2.	Recall The maximum value of Probability of Type I Error will be always	
	a. $=\alpha\%$	b. $<\alpha\%$
	c. $\leq\alpha\%$	d. $>\alpha$
3.	If β is the probability of Type II error, then name (1-β) is, select the correct option.	
	a. Type I error	b. Level of significance
	c. Power of the test	d. Critical Region
4.	State that a test is One tail or two tail depends on	
	a. Null hypothesis	b. Alternative Hypothesis
	c. Sample values	d. None of these
5.	If X and Y are two independent chi square variates with v_1 and v_2 d.f respectively, then recognize the distribution of$(X/v_1)/(Y/v_2)$	
	a. Chi-square distribution	b. t-distribution
	c. F-distribution	d. Z distribution
6.	For testing equality of mean, when population variance is known we use, select the correct option.	
	a. Z statistic	b. t statistic
	c. F statistic	d. Chi square statistic
7.	In a study, subjects are randomly assigned to one of three groups: control, experimental A, or experimental B. After treatment, the mean scores for the three groups are compared. Identify the appropriate statistical test for comparing these means.	
	a. Correlation Coefficient	b. T-test
	c. F-test	d. ANOVA

8.	In a large population of adults, 45% have a post secondary degree.If people are selected at random from this population, enumerate the probability that the third person selected is the first one that has a post secondary degree?	
	a. 0.1361	b. 0.1254
	c. 0.1567	d. None of these.
9.	When conducting an ANOVA, F-data will always fall within what range? Select the correct answer.	
	a. 0 to infinite	b. 0 to 5
	c. less than 0	d. 0 to 1
10.	If SSE=200 and df(Error)=10, enumerate the value of mean square error.	
	a. A. 20	b. B. 2000
	C. 200	c. D. 100
11.	For a treatment A, it is found that MSA=25 and degrees of freedom = 2, then enumerate the Sum of squares of treatment A.	
	a. 12.5	b. 50
	c. 25	d. d. 100
12.	A result is called “statistically significant” whenever _____. Select the correct answer.	
	a. The null hypothesis is true.	b. The alternative hypothesis is true.
	c. The p-value is less or equal to the significance level.	d. The p-value is larger than the significance level.
13	If a hypothesis test were conducted using $\alpha = 0.05$, enumerate for which of the following p-values would the null hypothesis be rejected.	
	a. 0.1	b. 0.041
	c. 0.055	d. 0.060
14.	The owner of travel agency would like to determine whether the mean age of the agency's customers is over 24. If so, he plans to alter the destination of their special cruises and tours. If he concludes the mean age is over 24 when it is not, he makes _____. select the correct option.	
	a. Type I error	b. Type II error
	c. No error	d. None of these.
15.	For one sample mean test with unknown variance we should use _____. select the test	
	a. Z-test	b. Chi-square test
	c. F-test	d. t-test
16.	If population standard deviation is known and $n>30$ then enumerate the appropriate test statistics mean comparison.	
	a. t-test	b. Z-test

	c. F-test	d. Chi-square test
17.	Identify which of the following formulas is used to calculate the F statistic for a one-way ANOVA experiment?	
	a. SSB/SSW	b. MSW/MSB
	c. SSW/SSB	d. MSB/MSW
18.	The value that separates a rejection region from an acceptance region is called a _____. Select the correct option.	
	a. Parameter	a. Critical Value
	b. Significance level	c. None of these
19.	which of the following is type II error? select the correct option.	
	a. The error of accepting H_0 when H_0 is true	b. The error of rejecting H_0 when H_0 is false
	c. The error of accepting H_0 when H_0 is false	d. The error of rejecting H_0 when H_0 is true
20.	The _____ sum of squares measures the variability of the observed values around their respective treatment means. Select the correct answer:	
	a. treatment	b. error
	c. interaction	d. total
21.	In a study, subjects are randomly assigned to one of three groups: control, experimental A, or experimental B. After treatment, the mean scores for the three groups are compared. Select the appropriate statistical test for comparing these means.	
	a.The correlation coefficient	b. chi square
	d. the t-test	c.the analysis of variance
22.	In one-way ANOVA, Select which of the following is used within the F-ratio as a measurement of the variance of individual observations?	
	a. SSTR	b. MSTR
	c. SSE	d. none of these
23.	When conducting a one-way ANOVA, the _____ between-treatment variability is when compared to the within-treatment variability, the _____ value of F-DATA will be tending to be. Select the correct answer.	
	a. smaller, larger	b. smaller, smaller
	c. larger, larger	d. smaller, more random
24.	When conducting an ANOVA, F-DATA will always fall within what range? Select the correct answer.	
	a. between negative infinity and infinity	b. between 0 and 1
	c. between 0 and infinity	d. between 1 and infinity

25.	If F-DATA= 0.9, the result is statistically significant _____. Select the correct answer.	
	a. Always	b. Sometimes
	c. Never	d. not possible to conclude
26.	You obtained a significant test statistic when comparing three treatments in a one-way ANOVA. In words, select the alternative hypothesis.	
	a. All three treatments have different effects on the mean response.	b. Exactly two of the three treatments have the same effect on the mean response.
	c. At least two treatments are different from each other in terms of their effect on the mean response.	d. None of the above
27.	ANOVA was used to test the outcomes of three drug treatments. Each drug was given to 20 individuals. The MSE for this analysis was 16. Enumerate the standard deviation for all 60 individuals sampled for this study?	
	a. 6.928	b. 48
	c. 16	d. 4
28.	In a one-way ANOVA, if the computed F value exceeds the critical F value, Select the decision which is made regarding the null hypothesis.	
	a. Reject H_0 since there is evidence of a treatment effect.	b. Do not reject H_0 since there is no evidence of a difference
	c. Reject H_0 since there is evidence that all means differ.	d. Do not reject H_0 because a mistake has been made.
29.	Select which of the following ANOVA components are not additive?	
	a. Sum of square	b. Mean sum of square
	c. degrees of freedom	d. all the above are additive
30.	A Type I error occurs. select the correct option.	
	a. reject null hypothesis when it is true	b. Reject alternative hypothesis when it is true
	c. Accept null hypothesis when it is false	d. Accept alternative hypothesis when it is false
31.	A hypothesis can be classified as_____. Select the correct option.	
	a. Null	b. simple
	c. Composite	d. all the above
32.	For testing of hypothesis critical region is also known as _____.Select the correct option.	
	a. confidence region	b. acceptance region
	c. rejection region	d. none of the above
33.	The chance of rejecting of a true hypothesis decreases when sample size is _____. Select the correct option.	

	a. increases	b. decreases
	c. c. constant	d. both (a) and (b)
34.	Select the degrees of freedom for the test statistic in one sample t-test.	
	a. 1	b. n
	c. n-1	d. 0
35.	The number of independent values in a set of values is called_____. Select the correct option.	
	a. Test statistic	b. degrees of freedom
	c. level of significance	d. level of confidence
36	A Z-test for a population mean is generally used when, select the correct option.	
	a. Population variance is unknown and sample size is small	b. Population variance is known or the sample size is large
	c. Sample size is always less than 30	d. Population follows a Poisson distribution only
37	To test a hypothesis about a single population variance, the appropriate test statistic is, elect the correct option	
	a. Z	b.t
	c. $\chi 2$(Chi-Square)	d. FFF
38	In a one-way ANOVA test, the null hypothesis is rejected when, elect the correct option	
	a. $F_{cal} < F_{tab}$	b. $F_{cal} = F_{tab}$
	c. $F_{cal} > F_{tab}$	d. The sample means are equal

Short Type Questions:

1	Explain the effect of sample size on the confidence interval in hypothesis testing.
2	Discuss type I error and type II error in the context of hypothesis testing.
3	Discuss one tailed and two tailed test.
4	Discuss critical region.
5	You represent a chemical company that is being sued for paint damage to automobiles. You want to support the claim that the mean repair cost per automobile is not $650. Identify the null and alternative hypotheses.
6	A government association claims that 44% of adults in the United States do volunteer work. You work for a volunteer organization and are asked to test this claim. You find that in a random sample of 1165 adults, 556 do volunteer work. At alpha = 0.05, do you conclude that you have enough evidence to reject the association's claim.

7	A random sample of size 20 from a Normal population gives a sample mean 42 and sample standard deviation 6. Estimate the value of test statistics if the population mean is 44 using appropriate test statistics.
8	A random sample of size 40 from a Normal population gives a sample mean 24 and sample standard deviation 10. Estimate the value of test statistics if the population mean is 26 using appropriate test statistics.
9	In order to test whether a coin is perfect, the coin is tossed 5 times. The null hypothesis of perfectness is rejected if and only if more than 4 heads are obtained. Estimate the probability of type I error.
10	In your work for a national health organization, you are asked to monitor the amount of sodium in a certain brand of cereal. You find that a random sample of 82 cereal servings has a mean sodium content of 232 milligrams. The population standard deviation is known to be 10 milligrams. Identify the null and alternative hypothesis.
11	A manufacturer claims that **60%** of his products are of superior quality. In a random sample of 400 products, 220 are found to be of superior quality. Test the claim at the **5% level of significance**.
12	A sample of size 30 from a normal population has a sample mean of 50 and a sample variance of 25. Test the hypothesis that the population mean is 52 at a 5% significance level.
13	The mean lifetime of a sample of 49 electric bulbs is found to be 980 hours. Assuming the population standard deviation to be 70 hours, test at the **5% level** whether the mean lifetime of the bulbs is **less than 1000 hours**.
14	Explain the assumptions that are necessary for performing the t-test for a population mean when the population variance is unknown.
15	A hospital claims that 90% of patients recover after treatment. Out of 150 patients, 126 recovered. Examine the claim at the 1% significance level.
16	The scores obtained by students taught using three different teaching methods are: **Method I Method II Method III** 70 75 80 72 74 79 71 76 81 Examine, using **one-way ANOVA**, the sum of square between(SSB).
17	The scores obtained by students taught using three different teaching methods are: **Method I Method II Method III** 70 75 80 72 74 79 71 76 81

	Calculate, using **one-way ANOVA**, the Sum of square within(SSW).
18	The lifetimes (in hours) of batteries produced by three machines are: **Machine A Machine B Machine C** 95 100 90 98 102 92 97 101 91 Using **one-way ANOVA**, calculate Mean square between(MSB).

Long Type Questions:

1	Suppose you start up a company that has developed a drug that is supposed to increase IQ. You know that the standard deviation of IQ in the general population is 15. You test your drug on 36 patients and obtain a mean IQ of 97.65. Using an alpha value of 0.05, evaluate if this IQ significantly different than the population mean of 100.
2	The purpose of a study by Luglie was to investigate the oral status of a group of patients diagnosed with thalassemia major (TM). One of the outcome measure s was the decayed, missing, filled teeth index (DMFT). In a sample of 18 patients, the mean DMFT index value was 10.3 with standard deviation of 7.3. Evaluate if this sufficient evidence to allow us to conclude that the mean DMFT index is greater than 9 in a population of similar subjects? Let $\alpha = 0.1$
3	Escobar performed a study to validate a translated version of the Western Ontario and McMaster University index (WOMAC) questionnaire used with Spanish-speaking patient s with hip or knee osteoarthritis. For the 76 women classified with sever hip pain. The WOMAC mean function score was 70.7 with standard deviation of 14.6, we wish to know if we may conclude that the mean function score for a population of similar women subjects with sever hip pain is less than 75. Let $\alpha = 0.01$. Evaluate the hypothesis.
4	Among 157 African-American men, the mean systolic blood pressure was 146 mm Hg with a standard deviation of 27. We wish to know if on the basis of these data, we may conclude that the mean systolic blood pressure for a population of African-American is greater than 140. Use $\alpha = 0.01$. Evaluate the hypothesis.
5	A dice is thrown 400 times and 'four' resulted 60 times. Do the data justify that the hypothesis of an unbiased dice.
6	A sample of nine plastic nuts yielded an average diameter of 3.1 cm with sample standard deviation of 1.0 cm. It is assumed from design and manufacturing requirements that the population mean of nuts is 4.0 cm. Evaluate the mean diameter of plastic nuts being produced.

7 A fertilizer mixing machine is set to give 12 kg of nitrate for every quintal bag of fertilizer. Ten 100 kg bags are examined. The percentage of nitrate are: 14, 11, 13, 12, 13, 12, 13, 11, 13, 12. Evaluate if there is a reason to believe that the machine is defective? Critical value for t-distribution for 9 d.f. is 2.262.

8 A certain stimulus administered to each of 12 patients resulted in the following changes in blood pressure: 5,2,8,-1,3,0,-2,1,5,0,4,6
Can it be concluded that the stimulus will in general be accompanied by an increase in blood pressure?(Given for 11 df $t_{0.05}=2.2$)

9 The following table shows the retail prices (Rs. Per kg) of a commodity in some shops selected at random in four cities:

cities			
A	B	C	D
34	29	27	34
37	33	29	36
32	30	31	38
33	34	28	35

Carry out the ANOVA to test the significance of the difference between prices of the commodity in the four cities. [given $F_{.05}=3.49$ for (3,12) degrees of freedom.

10 Test ANOVA for the following one way classified data and comment:

Machine I	Machine 2	Machine 3
2.0	1.8	3.0
2.2	2.2	2.8
1.7	2.0	3.2

[Given $F_{.05}=5.14$ for (2,6) degrees of freedom]

11 A sample of 50 observations from a normal population yielded a sample mean of 75 and a standard deviation of 10. Test the hypothesis if the population mean is 78 at 5% significance level.Clearly explain the null and alternative hypotheses, the test statistic used, and your conclusion.

12 In a factory, a machine produces a large number of items. It is known from past records that the machine produces 2% defective items. A new batch is produced, and a random sample is tested. One defective item is found.Test, at the 1% level of significance, whether the defect rate has increased from the historical rate of 2%. Clearly explain the null and alternative hypotheses, the test statistic used, and your conclusion

13 A manufacturer claims that the average life of his batteries is at least 100 hours. A sample of 25 batteries shows a mean life of 96 hours with a standard deviation of 10 hours. Test the manufacturer's claim at the 5% significance level.
[Use the appropriate test and clearly explain your hypotheses and conclusion.]

14 Define the following terms with suitable examples:
(a) Level of Significance
(b) Type I and Type II Errors
Also, explain the concept of *p-value* in hypothesis testing.

15 A company manufacturing LED bulbs claims that the average lifetime of its bulbs is 1500 hours. A random sample of 64 bulbs gives a mean lifetime of 1460 hours. The population standard deviation is known to be 120 hours. Illustrate the company's claim at the 5% level of significance.

16	A nutritionist claims that the average daily protein intake of college students is 65 g. A random sample of 20 students gives the following summary: • Sample mean = 61.8 g • Sample standard deviation = 6.5 g Illustrate the nutritionist's claim at the 5% level of significance, assuming normality.[$Given: t_{0.025,19} = 2.093$]
17	The average monthly electricity bill in a city is claimed to be ₹1800. A random sample of 100 households gives a mean bill of ₹1745. The population standard deviation is ₹250.At the 1% significance level, Explain whether the average monthly bill differs from ₹1800.
18	Three brands of batteries are tested to compare their average lifetimes (in hours). **Brand A** **Brand B** **Brand C** 90 95 100 92 96 101 91 97 102 93 98 99 Using one-way ANOVA, Illustrate whether the average lifetimes of the three brands are equal at the 5% significance level. **Given:** $F_{\{0.05\}(2,9)} = 4.26$